(http://mng.bz/Zq1P) and CI/CD and then was extended to security (http://mng.bz/RxdK). Nowadays, this term is also something marketing departments like to apply in the context of observability.

The concept of shift left (see figure 9.1) is to enable developers to do things that were traditionally testers' or operation folks' tasks and responsibilities. In the "good old days," and in rather conventional environments, developers still mainly focus on one thing: writing code—maybe with a bit of unit testing as well. At the end of the day, developers "throw their work over the wall" to be tested; packaged (e.g., into a WAR file); and, finally, deployed and operated. All of these steps are carried out by roles other than developers, such as release engineers, QA testers, or operators. If you think about the software development and operations process as a continuum, with the leftmost side representing developers churning out source code and the rightmost side representing operators deploying a binary, you can picture what *shift left* means.

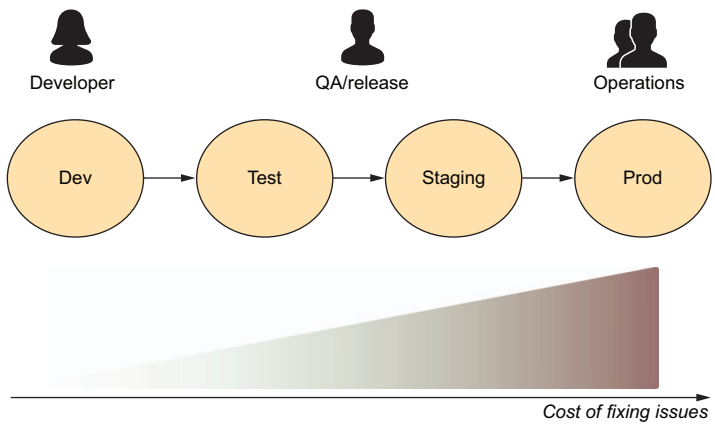


Figure 9.1 Shift left

Applying the concept of shift left to observability, *developer observability* means using telemetry signals to equip developers with actionable insights to accelerate development, debug code, or understand the impact of a new feature in terms of resource usage. This results in shortening the feedback loop, resulting in lower overall costs to identify and fix issues. For example, rather than a developer needing to wait four weeks to find out if the code passes a test suite or crashes in the prod environment, they can, and oftentimes now must, run the test suite themselves in the process of sending in a pull or merge request.

In this chapter, we will first discuss a specific developer observability method called *continuous profiling* and how it can help you understand the resource usage (e.g., CPU and memory) of the code you write, even across different versions of your source code. Then, we will have a look at developer productivity based on observability tooling deeply integrated into editors and IDEs that give you, as a developer, insights into